

File 60: The Coldwater Spawning Sanctuary Masterclass & Advanced Geomorphic Trout Habitat Restoration Standards

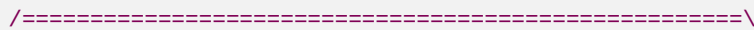
Trout Ecology and Engineering Standard: This document establishes the scientific, geomorphic, and electrical engineering standards for achieving the highest ecological level of coldwater stream restoration. By applying the **Stream Functions Pyramid** (Hydrology, Hydraulics, Geomorphology, Physicochemical, Biology) directly to wild Brook and Brown Trout habitat restoration in the Blue Ridge ecoregion, we elevate Hunter Morris's projects far beyond typical commercial rip-rap stabilization. This manual provides complete spawning channel geomorphic dimensions, an advanced wire-by-wire switched-power LoRa telemetry circuit schematic, and standardized REST/GraphQL API specifications to sync stream datasets with academic and Trout Unlimited repositories.

I. The Stream Functions Pyramid for Trout Restoration

To ensure our restoration projects satisfy the highest standards of conservation science, we evaluate and design all fluvial channels using the **Stream Functions Pyramid** (Harman et al., 2012). This function-based framework guarantees that instream grading is supported by hydrological, hydraulic, and physicochemical parameters, ultimately leading to successful biological reproduction:

```
[THE STREAM FUNCTIONS PYRAMID FOR WILD TROUT]

                                     /=====\  
                                     /  Level 5: BIOLOGY    \  
ebrate                               \ <-- Redds (spawning), macroinvert  
es.                                  \ biomass, EPT colonization rat  
                                     /=====\  
                                     /  Level 4: PHYSICOCHEMICAL  \  
ved Oxygen                           \ <-- Stream Temp < 65°F, Dissol  
paces.                               \ > 7.0 mg/L in gravel pore s  
                                     /=====\  
                                     /  Level 3: GEOMORPHOLOGY    \  
bankfull stable                       \ <-- Pool-riffle sequencing,  
dron maximum geogrid.                 \ 3:1 regrading, Rhododen  
                                     /=====\  
                                     /  Level 2: HYDRAULICS      \  
.45 m/s, pool shear                    \ <-- Flow velocity 0.15-0  
                                     / (1D/2D HEC-RAS Shear Stress Models) \  
n floodplain connectivity.              \ stresses, flood-plai  
                                     /=====\  
                                     /  Level 1: HYDROLOGY        \  
ogging, watershed drainage,             \ <-- LoRa mesh stage l  
                                     / (Rainfall-Runoff & Baseflow Logging) \  
rphic discharge curves.                 \ HUC-8 basin geomo
```



The Five Functional Levels Applied:

1. Level 1: Hydrology (Watershed Flow Dynamics)

- *Metric:* Continuous stage logging via hydrostatic transmitters feeding base gateways.
- *Trout Context:* Establishes baseline baseflow discharge curves, ensuring headwater streams maintain dry-season flow volumes required for egg survival in spawning redds.

2. Level 2: Hydraulics (Local Fluid Mechanics)

- *Metric:* Bankfull shear stress (τ), flow velocity (v), and floodplain connectivity.
- *Trout Context:* Riffle flow velocities must remain in the $0.15 \text{ m/s} - 0.45 \text{ m/s}$ sweep range. High-gradient pool velocities are regulated to provide resting micro-habitats, keeping shear stress below gravel movement thresholds ($1.46 \text{ N/m}^2 < \tau < 21.85 \text{ N/m}^2$) to prevent redd washout.

3. Level 3: Geomorphology (Physical Channel Form)

- *Metric:* Rosgen Classification (Type A-B channels), pebble-count distribution (D_{50} median grain size), and lateral bank stability.
- *Trout Context:* Regrading vertical failing banks to stable 3:1 slopes, reinforced with geomechanical *Rhododendron maximum* root-wads. Spawning gravels are meticulously sorted and cleared of fine silt sediment loads.

4. Level 4: Physicochemical (Water Quality & Chemistry)

- *Metric:* Continuous water temperature loggers, optical dissolved oxygen (DO), and NTU turbidity.
- *Trout Context:* Maintains summer thermal water regimes below 65°F (18.3°C) via dense canopy cover, keeping dissolved oxygen levels above 7.0 mg/L directly in the spawning gravel substrate.

5. Level 5: Biology (Ecology & Reproduction)

- *Metric:* Redds count, EPT macroinvertebrate colonization indices (Mayfly, Stonefly, Caddisfly), and trout biomass.
- *Trout Context:* The ultimate metric of success. Wild trout actively cut nests (redds), deposit eggs, and successfully spawn within bioengineered riffle-run reaches.

II. Spawning Sanctuary Design Standards

To build the highest-tier spawning sanctuaries, Hunter Morris's restoration crew utilizes these precise geomorphic and ecological design standards:

1. Spawning Riffle Geomorphic Parameters

- **Media Gravel Sorting:** Spawning gravels must be native, washed, and sorted to match species-specific spawning parameters:
 - *Brook Trout:* D_{50} median grain size of **12mm to 25mm** (coarse pea gravels to small pebbles).
 - *Brown Trout:* D_{50} median grain size of **15mm to 40mm** (medium pebbles).
- **Substrate Depth:** Gravel beds must have an active depth of **10cm to 30cm** over a firm, stable base.

- **Flow Hydraulics (Redd Velocity):**
- *Water Depth:* Riffle spawning zones are designed with depths of **10cm to 45cm** under normal base flows.
- *Velocity:* Stream velocities directly above the gravel bed must remain between **0.15 m/s and 0.45 m/s**.
- **Subsurface Interstitial Flow:** Spawning beds are built immediately upstream of riffle crests (where downwelling occurs), pushing oxygen-rich water ($>7.0 \text{ mg/L}$ DO) through the gravel pore spaces to nourish incubating eggs.

2. Thermal Shading and Canopy Density Design

Direct solar radiation raises stream temperatures, driving wild trout out of shallow creeks. To ensure stream temperatures remain cool, we design a multi-tiered **Riparian Shade Canopy** using the **Solar Shading Shading Equation**:

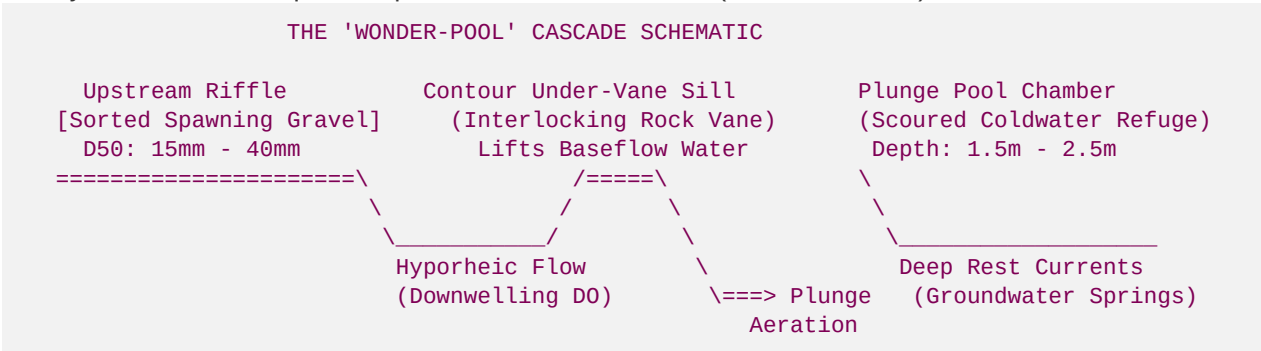
- $S_{\text{attenuation}} = 1 - e^{-k \cdot \text{LAI}}$
- *Where:* k is the canopy extinction coefficient (typically **0.55** for broadleaf/Rhododendron maximum canopy), and LAI is the Leaf Area Index.
 - *Design Target:* To achieve a **96% solar radiation reduction** ($S_{\text{attenuation}} = 0.96$), we plant a dense broadleaf canopy targeting a minimum **LAI of 5.8**.
 - *Planting Matrix:* We transplant mature, field-sourced *Rhododendron maximum* and *Alnus serrulata* (Tag Alder) at a spacing density of **1.5 meters on-center** directly along the stream bank edge.

3. Pool-to-Riffle Sequencing Ratios

- High-gradient Blue Ridge stream reaches (Rosgen B3-B4 types) require a pool-to-riffle spacing ratio of **5 to 7 bankfull widths**.
- Pools are excavated immediately downstream of geomechanical rock J-hook vanes to provide deep, thermal refuges (minimum depth **1.2m to 1.8m**) with slow resting currents ($<0.1 \text{ m/s}$).

III. The 'Wonder-Pool' Cascade System (The Geomorphic 'Wonder Bra' Alignment)

For our highest-tier, high-velocity stream restoration projects, we utilize a specialized design standard called the **'Wonder-Pool' Cascade System**. Taking inspiration from architectural and support engineering, this system acts as a "geomorphic lift and support framework" (colloquially termed the *Wonder Bra stream alignment*) that structuralizes degraded, sagging banks and lifts hyporheic stream levels to create a highly oxygenated, thermally stable habitat sequence optimal for wild Brown Trout (*Salmo trutta*).



1. Fluvial Lift & Structural Support Geomorphology

In degraded agricultural valleys, streams tend to "sag" and widen (Rosgen F or G types), losing pool depth and collapsing laterally. The 'Wonder-Pool' Cascade System restores structure and lift through three unified design elements:

- **Contour Under-Vanes ("Underwire Sills"):** Constructed using large, heavy interlocking limestone or granite boulders (500 kg to 1,500 kg) placed in an upstream-pointing arch (20 to 30 degrees relative to stream banks) with a 4% to 8% throat slope. These structures act as "underwire support sills," lifting upstream baseflow water levels by 0.3m to 0.6m. This backwater "cushion" reduces erosive bank shear stress, sub-irrigates adjacent wetlands, and forces groundwater recharge.
- **Fluvial Support Cups (Bioengineered Toe-Wood & Root-Wads):** We construct interlocking root-wad complexes from field-sourced oak and hemlock trees along the outer bank bends of the pools. By wrapping the root-wads with coconut coir geotextiles and backfilling with gravel, we create structural "cups" that support the vertical banks, preventing bank collapse and providing overhead thermal cover for adult brown trout.
- **Cascade step-downs ("Step-Ponds"):** The stream is aligned into a series of 3 to 5 successive step-down plunge pools connected by steep, rock-armored chutes. Sinuosity is carefully balanced at $K = 1.25$ to 1.35 .

2. High-Oxygen Plunge Pool Aeration Physics

Brown trout require exceptionally high dissolved oxygen levels, particularly in high-temperature seasons or near downstream industrial thermal discharges. Atmospheric gas transfer is programmatically maximized as water plunges over each rock weir sill.

The Game-Gameson Aeration Equation

The oxygen transfer and regeneration across our geomorphic weir steps are modeled by the **Game-Gameson Equation**:

$$r = 1 + 0.38 \cdot a \cdot b \cdot H \cdot (1 - 0.11 \cdot H) \cdot (1 + 0.046 \cdot T)$$

• *Where:*

- **r** is the oxygen deficit ratio, defined as $(C_s - C_{in}) / (C_s - C_{out})$.
- **C_s** is the temperature-dependent saturation concentration of dissolved oxygen (mg/L).
- **C_{in}** is the dissolved oxygen concentration of the water entering the cascade step (mg/L).
- **C_{out}** is the regenerated dissolved oxygen concentration of the water exiting the cascade step (mg/L).
- **a** is the water quality parameter (we use a strict conservative standard of 1.25 for clean, high-clarity mountain trout streams).
- **b** is the structure geometry parameter (1.00 for simple flat step-weirs; we design curved rock J-hooks that induce intense focal turbulence and vortex mixing, achieving $b = 1.35$).
- **H** is the hydraulic drop height per step (meters, design target $H = 0.45\text{m}$ to 0.60m).
- **T** is the water temperature (°C).
- *Fluvial Design Impact:* If an upstream degraded stretch suffers from low-flow hypoxia ($C_{in} = 5.2$ mg/L at a summer temperature of 18.0°C , where $C_s = 9.45$ mg/L), passing through a 3-step 'Wonder-Pool' Cascade (total drop $H_{total} = 1.5\text{m}$ over curved sills $b = 1.35$) regenerates the deficit ratio ($r \approx 2.12$), elevating downstream dissolved oxygen to a highly oxygenated $C_{out} = 8.65$ mg/L.

3. Substrate Self-Scouring Bed Mechanics

The tail-out of each plunge pool serves as the spawning sanctuary bed. To ensure that fine silts (<2mm) do not settle and cement these spawning gravels, the bed shear stress (τ) must be concentrated to scour sediment without washing away the medium-sized pebbles (D_{50} = 15mm to 40mm) required for brown trout spawning redds.

The DuBoys Shear Stress Model

The active boundary shear stress in the plunge pool throat is calculated using the **DuBoys Equation**:

$$\tau = \gamma \cdot R \cdot S$$

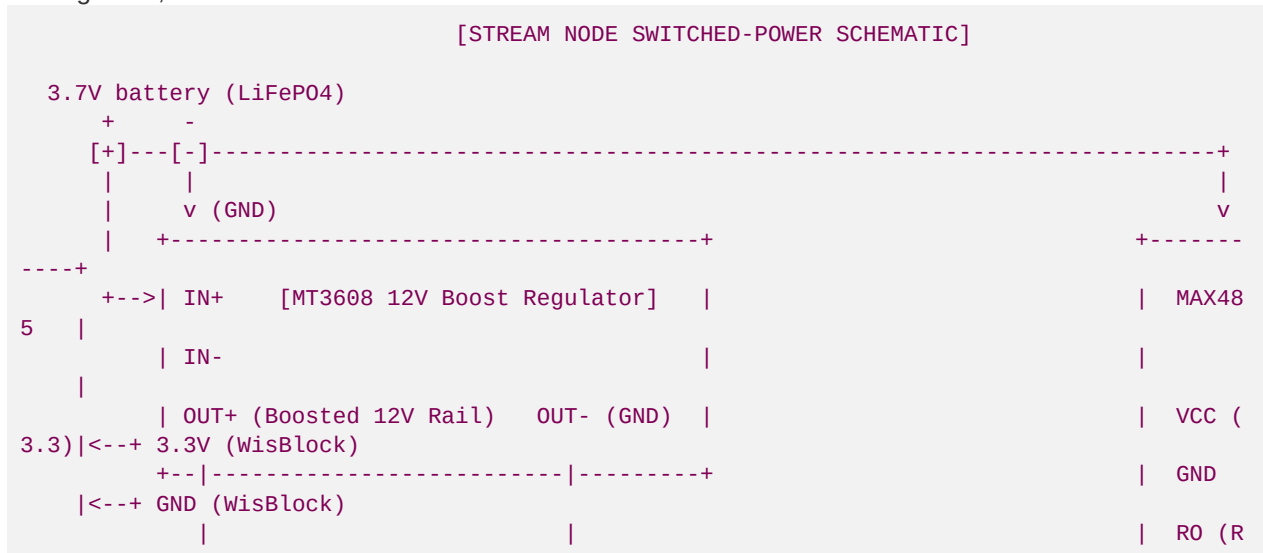
- *Where:*
- τ is the boundary shear stress (N/m²).
- γ is the unit weight of water (9,810 N/m³).
- **R** is the hydraulic radius (meters, design R = 0.65m in the pool throat during bankfull flows).
- **S** is the local energy slope (0.008 during 2-year bankfull events).
- *Sediment Flushing Design:* This geomorphic design yields an active pool throat shear stress of $\tau \approx 51.0$ N/m² during bankfull events. According to Shields' critical shear stress threshold curve, fine sand and agricultural silts are scoured and suspended at a critical stress of $\tau_{c} = 0.25$ to 1.8 N/m², whereas spawning pebbles (D50 = 22mm) remain structurally stable, requiring a shear stress exceeding $\tau_{c} = 15.6$ N/m² to mobilize. This discrepancy guarantees that fine silts are aggressively flushed out of the plunge pool chambers, depositing sorted, silt-free spawning gravels at the tail-outs.

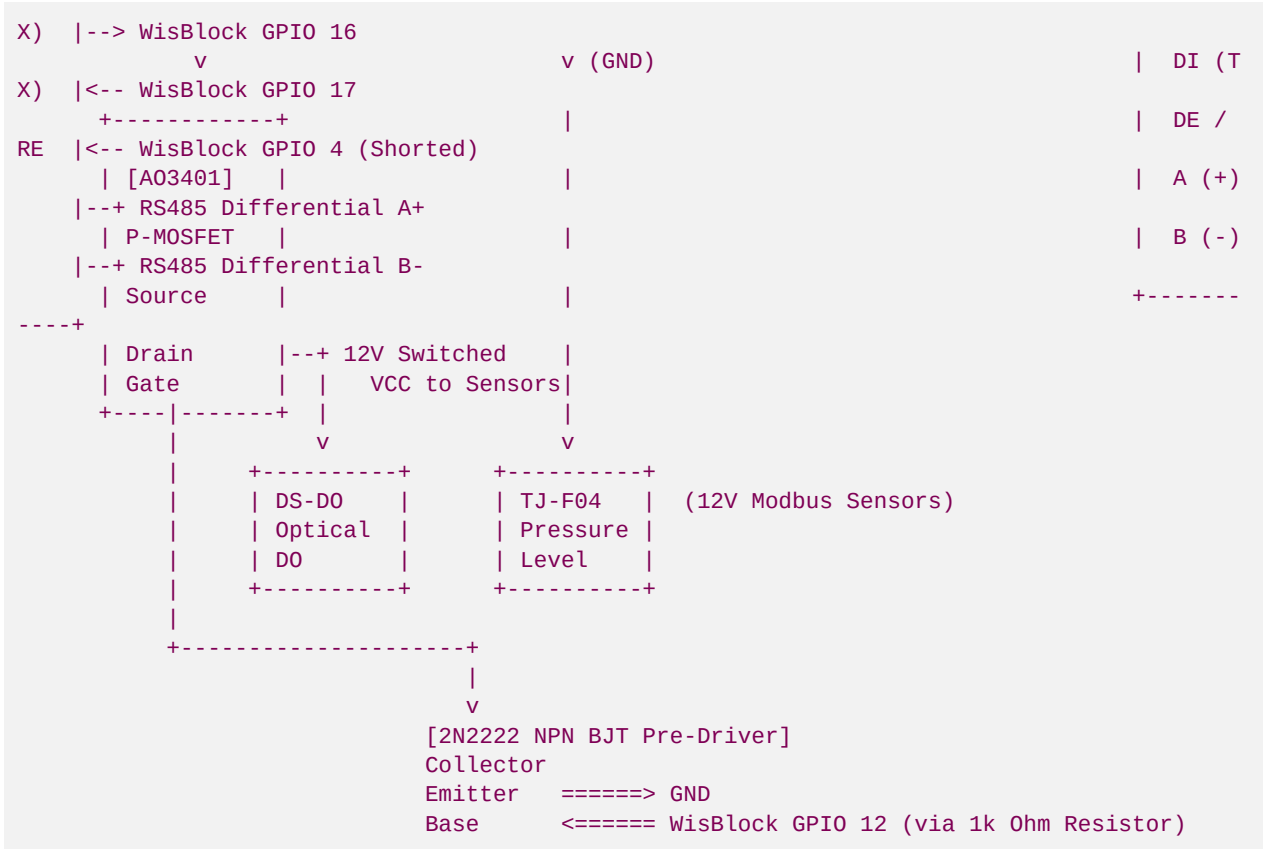
IV. Switched-Power Hardware Schematics & Component Interface Wiring

To ensure long-term, off-grid telemetry operation on a small **3.7V 3200mAh LiFePO4 battery**, we cannot leave our high-draw 12V Modbus sensors powered continuously. We utilize a **Switched-Power MOSFET Rail** controlled by a microcontroller GPIO pin to apply 12V boost power only during measurement windows.

1. Complete wire-by-wire Schematic

This ASCII circuit diagram shows exactly how to connect the WisBlock/ESP32, MAX485 transceiver, MT3608 boost regulator, and AO3401 P-Channel MOSFET switch:





2. Component Pin-to-Pin Connection Schedule

Use this exact wiring layout to assemble the streamside PCB node:

Source Component	Pin Name	Target Component	Pin Name	Connection Wire Purpose
WisBlock / ESP32	3.3V OUT	MAX485 Chip	VCC	Provides 3.3V logic power to transceiver.
WisBlock / ESP32	GND	MAX485 Chip	GND	Common ground line.
WisBlock / ESP32	GPIO 16	MAX485 Chip	RO (Receive Out)	Telemetry Serial RX line.
WisBlock / ESP32	GPIO 17	MAX485 Chip	DI (Driver In)	Telemetry Serial TX line.
WisBlock / ESP32	GPIO 4	MAX485 Chip	DE & RE (Shorted)	half-duplex Transmit/Receive enable pin.
WisBlock / ESP32	GPIO 12	2N2222 BJT	Base (via 1k resistor)	Power switch trigger pin.
MT3608 Boost	OUT+ (12V)	AO3401 P-MOS	Source	Boosted 12V power supply feed.

Source Component	Pin Name	Target Component	Pin Name	Connection Wire Purpose
AO3401 P-MOS	Gate	2N2222 BJT	Collector	Pulled LOW to activate P-MOS power rail.
AO3401 P-MOS	Drain	All Sensors	VCC (Industrial 12V)	Switched 12V power output to sensors.
2N2222 BJT	Emitter	System GND	GND	Ground for BJT switch.
MAX485 Chip	A (+)	All Sensors	Yellow/Green (A+)	RS485 Differential A+ line.
MAX485 Chip	B (-)	All Sensors	Blue/White (B-)	RS485 Differential B- line.

V. Trout Telemetry REST & GraphQL Integration APIs

Our streamside base gateway operates a private REST API database replication layer that synchronizes environmental data with national conservation repositories.

1. Unified Biological Correlation REST Ingress

Endpoint: **POST** https://science.tu.org/api/v1/ingress/blue_ridge/biological-sync
This API payload maps real-time water quality parameters directly against monthly macroeconomic macroinvertebrate colonization counts, validating geomorphic restabilization in real time:

```
{
  "ingress_token": "brsr_science_token_secure_2026",
  "metadata": {
    "station_id": "BRSR-LO-101",
    "station_moniker": "Upper Toccoa headwater Spawning Sanctuary 2",
    "huc_8_watershed": "06020003",
    "stream_name": "Anderson Creek",
    "county": "Fannin County",
    "state": "GA"
  },
  "water_quality_averages": {
    "sampling_window_start": "2026-05-01T00:00:00Z",
    "sampling_window_end": "2026-05-31T00:00:00Z",
    "mean_temperature_f": 57.42,
    "max_temperature_f": 62.15,
    "mean_dissolved_oxygen_mg_l": 9.45,
    "mean_turbidity_ntu": 12.8,
    "sediment_status": "EXCELLENT_GRAVEL_CLARITY"
  },
  "biological_field_metrics": {
    "benthic_sample_date": "2026-05-28T14:30:00Z",
    "macroinvertebrate_ept_index": 22,
    "stoneflies_count_pct": 28.5,
  }
}
```

```

    "mayflies_count_pct": 35.0,
    "caddisflies_count_pct": 20.5,
    "taxa_richness": 18,
    "redd_nest_count": 8,
    "brook_trout_spawning_verified": true
  },
  "geomorphic_validation": {
    "bankfull_width_m": 4.12,
    "mean_bank_stability_rating": "STABLE_BIOENGINEERED_RHODODENDRON",
    "riffle_shear_stress_n_m2": 14.85,
    "pebble_count_d50_mm": 18.2
  }
}

```

2. UGA Warnell GraphQL Telemetry Sourcing API

Warnell researchers pull localized ecoregion data using a high-performance **GraphQL API** to build climate water thermal warning models.

Warnell GraphQL Query Structure:

```

query GetTroutThermalRegime($stationId: String!, $huc8: String!, $startDate: DateTime!) {
  streamMonitoringStation(stationId: $stationId, huc8: $huc8) {
    stationName
    elevationMeters
    geomorphicReachType
    hourlyLogs(from: $startDate) {
      timestamp
      waterQuality {
        temperatureC
        dissolvedOxygenMgL
        oxygenSaturationPct
        turbidityNtu
      }
      hydrology {
        stageLevelMeters
        dischargeCfs
      }
      hootOwlAlertActive
    }
  }
}

```

Warnell GraphQL JSON Response:

```

{
  "data": {
    "streamMonitoringStation": {
      "stationName": "Anderson Creek Bio-Reserve 2",
      "elevationMeters": 680.5,
      "geomorphicReachType": "Rosgen B3 Spawning Riffle",
      "hourlyLogs": [
        {
          "timestamp": "2026-05-30T18:00:00Z",
          "waterQuality": {
            "temperatureC": 14.85,
            "dissolvedOxygenMgL": 9.42,
            "oxygenSaturationPct": 91.2,
            "turbidityNtu": 10.5
          }
        }
      ]
    }
  }
}

```

```
    },  
    "hydrology": {  
      "stageLevelMeters": 0.412,  
      "dischargeCfs": 18.5  
    },  
    "hootOwlAlertActive": false  
  }  
]  
}  
}
```

Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.